

Two-Axis Robotic Arm

Yunseong Cho - 934546776

Koh Suzaki - 934573008

DeAngelo Joshua Ola - 934621759

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Project Plan

Yunseong Cho: Just as I have always been interested in electrical engineering, I have also been very interested in robotics. Robotics is a prime example of a convergent engineering field that combines mechanical and electrical engineering. Therefore, through this robotic arm project, I intend to learn how to control mechanical systems using electrical engineering. In particular, I am eager to gain a deep understanding of system control aspects, such as the ability to control the robotic arm's motors or sensors using electronic signals, and how to convert user input into digital signals to communicate with the system. Furthermore, I hope to gain experience in teamwork—specifically, effectively communicating with teammates to solve problems when engineering solutions are required, such as for customer requirements—and to gain firsthand experience in how engineers actually work in real-world industrial settings.

Koh Suzaki: I am interested in learning more about system design, interface definition, and hardware integration. By working on this project, I can develop skills in block diagram design, interface validation, power distribution, and documentation. This supports our design goals because the robotic arm must integrate multiple subsystems, including power, control, motors, and the user interface, into one complete and verifiable system.

DeAngelo Joshua Ola: I think that I am most interested in learning how mechanical systems can pair with electrical systems. For example with the robotic arm I want to see how its mechanics play a role. I have always been interested in robotics as it is such an interesting field. I think this project will definitely help me in helping me get a good understanding of robotics and the merging of mechanical engineering and electrical engineering.

Engineering Requirements

- 1. Customer Requirement: The system should be fast.**
Engineering Requirement: The system must draw with a speed of 4 inches per second.
- 2. Customer Requirement: The system must be accurate.**
Engineering Requirement: The system must draw a 10 inch straight line +/- .25 inch in all directions.
- 3. Customer Requirement: The system needs to be inexpensive and manageable to manufacture.**
Engineering Requirement: The system will use a SCARA topology, with two rotating joints to control arm actuation.
- 4. Customer Requirement: The system must have a commonly known interface.**
Engineering Requirement: The system input will be G-code commands. These commands must be made available within the Python or MATLAB GUI.
G0, G1, G90, G91, G20, G21, M2, M6.

5. **Customer Requirement: The system must have an emergency stop.**
Engineering Requirement: The system will stop moving in less than 1 second after user input and then return to normal operation when the emergency stop input is reset.
6. **Additional Customer Requirement: The system must allow the user to select the size of the drawing.**
Engineering Requirement: The system will draw a circle with a user-selectable radius of 1 in to 4 in within ± 0.25 in radial error.
7. **Additional Customer Requirement: The system must be portable and quick to set up.**
Engineering Requirement: The system will consist of a single assembled unit and will be ready for use within 1 minute after connection to a power source.

Top-Level Architecture Block Diagram

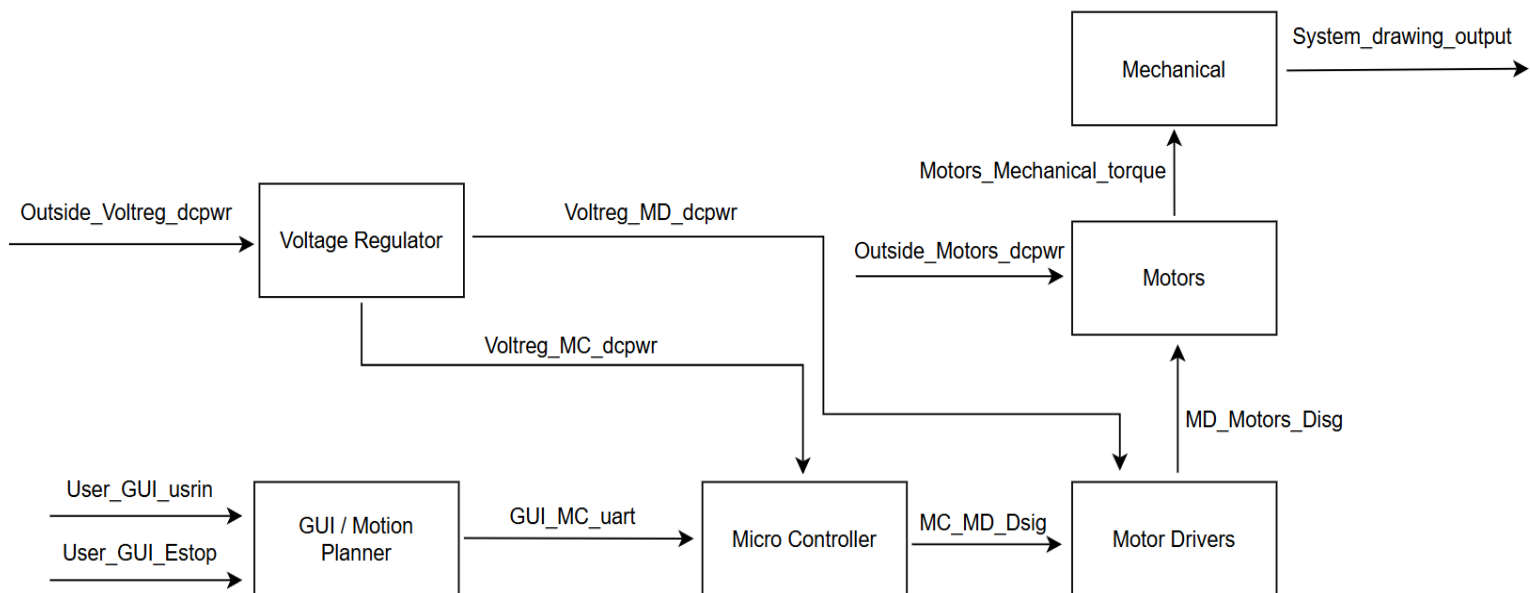


Fig 1. Top-Level Architecture Block Diagram.[1]

- Yunseong Cho: Voltage Regulator, GUI / Motion Planner
- Koh Suzaki: MicroController, Motor Drivers,
- DeAngelo Joshua Ola: Motor, Mechanical